



SporeScope — Imaging Rig

2025

A compact 16 × 15 × 18 in imaging rig built from 2020 aluminum extrusion and assembled with corner brackets and standard extrusion hardware. It holds 100 × 15 mm malt extract agar plates in fixed positions beneath a Raspberry Pi-controlled overhead camera, enabling consistent time-series image capture.

Aluminum Extrusion

Raspberry Pi

Backlit Imaging

Python

React



Bio Chart — Sensor & Imaging Chamber

2023

A compact sensor and imaging chamber built to collect repeatable data from biological samples. The enclosure contains sample shelves, sensors, controlled overhead lighting, and a camera for consistent image capture. A Raspberry Pi mounted externally connects to the internal components and coordinates data collection and transmission.

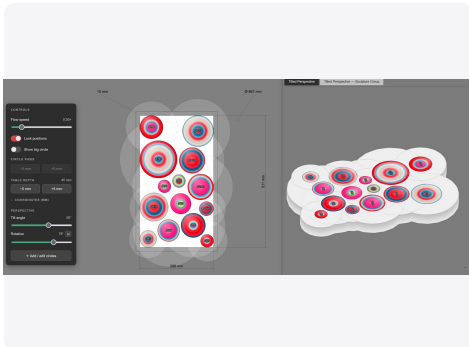
Raspberry Pi

Microcontrollers

Sensors

Python

React



Shared Signal

IN PROGRESS

An interactive sculpture system combining physical fabrication with embedded computing.

Fabrication

Raspberry Pi

LCD

JavaScript

SVG



3D-Printed Coffee Dosing Cone

FINISHED · IN USE

A flexible dosing cone made for a 58 mm portafilter. It grips the basket rim by friction, prevents coffee grounds from spilling, and removes easily with one hand.

CAD

3D Printing

TPU

Vlad Filitovich

Multidisciplinary R&D Engineer combining hardware prototyping, manufacturing processes, experimental design, and software development.

R&D PROJECTS

filitovich.com/rd.html



Laser Cutting & Engraving

ONGOING

An ongoing set of small parts cut and engraved on the laser — the kind of thing that takes ten minutes to draw and then quietly makes daily life work better.

Laser Cutting

Engraving

Plywood

Inkscape

Vector Design



Plant Imaging Rig

IN PROGRESS

A wall-mounted station for photographing a plant the same way, over and over, as it grows.

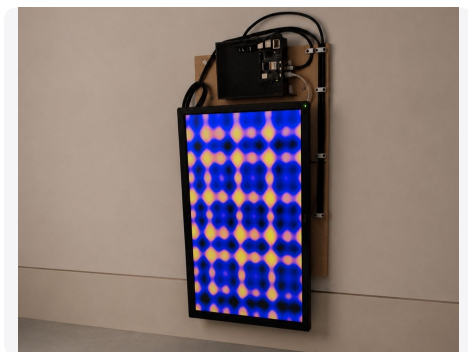
Raspberry Pi

Camera

Motor Control

Python

Capture Scheduling



Waiting-Room Display / Kiosk

IN PROGRESS

An unattended information display built from an older computer and an external monitor, wall-mounted in portrait orientation on a plywood backer with the machine and cabling mounted behind the screen.

Repurposed Hardware

Kiosk Mode

Wall Mount

GitHub Pages

HTML/CSS